
The Role of Individual Differences in the Study Abroad Context: Cognitive Capacity and Language Development During Short-Term Intensive Language Exposure

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While research suggests that study abroad (SA) benefits second language (L2) oral fluency, its benefits for other domains are less clear, especially for shorter-term programs, which are becoming more common. Additionally, studies investigating the relationship between cognitive capacity and benefits of SA report inconsistent patterns. In light of these gaps, this study investigated whether a 5-week intensive language experience abroad benefits L2 lexical and grammatical development and whether development is related to learners' cognitive capacity. Twenty-six L1 English-advanced L2 Spanish participants in a SA program in Spain completed pre-/post-SA grammaticality judgment (GJT) and lexical decision tasks (LDT). On the GJT, results showed higher accuracy and decreasing reaction times (RTs); specifically, accuracy increased on word order and number agreement items but not on gender agreement. RTs did not decrease for any of the three targets individually. For the LDT, performance on nonwords, but not words, showed improvement while RTs for both words and nonwords decreased. Overall, these gains following short-term experience abroad were independent of variation in cognitive capacity. This study makes a unique contribution toward understanding the roles of L2 proficiency, context, and individual differences in morphosyntactic and lexical development.

Keywords: advanced learners; intensive; study abroad; working memory; individual differences; Spanish

IT IS POPULARLY ASSUMED THAT STUDYING a second language (L2) abroad in an intensive

exposure context is superior to classroom instruction because it provides the optimal conditions for L2 learning to proceed more quickly and accurately. However, over 20 years of theoretical and empirical inquiry into the potential effects of an immersive L2-intensive context in adult second language acquisition (SLA) have not

rendered clear-cut findings (e.g., Collentine, 2009; Collentine & Freed, 2004; DeKeyser, 1991, 2007; Freed, 1990, 1995, 1998; Freed, Segalowitz, & Dewey, 2004; Lafford, 1995; Llanes, 2011; Muñoz, 2012; Pérez-Vidal, 2011, 2014; Sanz, 2014). In fact, the only clear consensus based on empirical evidence indicates that no “context of learning is uniformly superior to another for all students, at all levels of language learning, and for all language skills” (Collentine & Freed, 2004, p. 164).

The pedagogical and exposure contexts in which L2 learning takes place vary greatly but are minimally tied to (a) the quality and quantity of L2 input, with resulting implications for intake,¹ (b) the frequency of communicative opportunities for learners to practice and potentially automatize their L2 skills, and (c) the type of language instruction provided. For instance, whereas nonintensive classroom instruction is associated with learning under formal instructional conditions in an at home (AH) setting, L2 learning under more naturalistic conditions can occur in a domestic context via foreign language (one-way), bilingual (two-way), or indigenous immersion (Fortune & Tedick, 2008; Genesee, 1996; Tedick, Christian, & Fortune, 2011) as well as overseas in study abroad (SA) settings (Muñoz, 2012; Pérez-Vidal, 2014). In spite of differences in the learner populations they serve, L2 immersion at home and abroad are similar in the number of opportunities they offer students to hear, read, write, and speak the language, although there may be some qualitative differences in the type of interactions.

Nonetheless, not all SA programs are the same nor do they produce the same outcomes (Llanes, 2011), which may partially explain the inconclusive findings in SLA research (DeKeyser, 2014; Sanz, 2014). For example, SA programs can differ according to the duration of the program, students’ proficiency in the target language upon entry, the type and context of course work, the type of student housing, and the nature and extent of opportunities for guided/structured cultural interaction, reflection, and experiential learning, among other areas (see Engle & Engle, 2003). The first two factors—program duration and initial L2 proficiency—have been identified as particularly important for understanding the extent to which L2 immersion abroad facilitates L2 development, though both factors remain underinvestigated (DeKeyser, 2014; Llanes, 2011).

In terms of program duration, most existing studies have examined language gains over

semester, 6-month, or year-long stays abroad, with a few exceptions (e.g., Cubillos, Chieffo, & Fan, 2008; Llanes & Muñoz, 2009; Yager, 1998). However, the focus on longer-term programs does not align with the increasing popularity of shorter programs. According to the International Institute of Education (IIE, 2013), of the 283,223 U.S. college students who studied abroad during the 2011–2012 school year, 59% participated in short-term programs lasting 8 weeks or less, 38% in mid-length programs lasting one to two quarters or a full semester, and only 3% of students spent a year abroad. Thus, targeted investigations of the potential benefits of short-term programs are sorely needed.

Additionally, learners’ proficiency level in the target language prior to immersion, whether via intensive exposure abroad or at home, likely mediates development. While several studies show relatively more gains in oral skills for lower versus higher proficiency learners (e.g., Duperron & Overstreet, 2009; Llanes & Muñoz, 2009; Juan-Garau, 2014; Valls-Ferrer & Mora, 2014), DeKeyser (1991, 2007, 2010, 2014) has argued that learners need to possess a certain level of declarative language knowledge in order to maximize the potential to proceduralize their language skills during a study abroad experience. Given the paucity of research on L2 development in advanced learners in SA settings, more empirical evidence is needed to corroborate this claim.

Another area in need of further research is the effects of intensive exposure contexts in promoting L2 skills other than oral fluency, which is generally reported to show superior development in SA compared to AH contexts (e.g., Collentine & Freed, 2004; Freed, 1995; Freed et al., 2004; Llanes & Muñoz, 2009; Llanes, Tragant, & Serrano, 2011; Trenchs-Parera, 2009), though not for all measures of fluency (Freed, 1995; García-Amaya, 2009) and not in comparison to intensive domestic immersion settings (Freed et al., 2004; Serrano, Llanes, & Tragant, 2011). On the other hand, we know less about grammatical and lexical accuracy. While some studies have reported beneficial effects of SA on grammatical accuracy (e.g., Duperron, 2006; Guntermann, 1995; Isabelli & Nishida, 2005; Yager, 1998) or lexico-grammatical competence (Juan-Garau, Salazar-Noguera, & Prieto-Arrabz, 2014), other studies report no substantial differences between SA and AH contexts (e.g., DeKeyser, 1991, 2010; Isabelli-García, 2010), a limited impact on morphosyntactic system restructuring (Howard, 2001), and even superior gains for the formal

classroom AH setting (Collentine, 2004). Similarly, some studies report an association between superior gains in both lexicon size and accuracy as a function of SA experience (e.g., Dewey, 2008; Ife, Vives, & Boix, 2000; Milton & Meara, 1995), while others report either no learning effects or superior development in the AH setting (Collentine, 2004).

These contradictory findings may be linked to learner-internal individual differences (IDs) that likely interact with the learning context and, subsequently, with L2 learning outcomes (e.g., DeKeyser, 1991; Sanz, 2014). While psychosocial factors such as learner attitudes and motivation have been extensively studied in SA contexts (e.g., Allen, 2010; Freed, 1990; Hokanson, 2000; Isabelli-García, 2006; Larson-Hall & Dewey, 2012; Llanes et al., 2011; Serrano, Tragant, & Llanes, 2012; Yager, 1998), the relationship between cognitive capacity and L2 development has not figured prominently in the study of immersion or SA contexts. This stands in contrast to a prominent focus on investigations of the role of cognitive capacity in formal instructed L2 contexts (see Juffs & Harrington, 2011; Linck et al., 2013, for reviews). In fact, a search of studies that investigated cognitive ability in SA research returned only six published papers (LaBrozzi, 2012; O'Brien et al., 2007; Segalowitz & Freed, 2004; Sunderman & Kroll, 2009; Taguchi, 2008; Tokowicz, Michael, & Kroll, 2004). The small number of studies as well as the methodological variability among studies—including the operationalization of SA experience, learner proficiency level, and confounds between measures of cognitive ability administered in the L2 and L2 proficiency—prohibit reliable interpretation of findings.

To summarize, there are currently several gaps regarding our knowledge of the efficacy of intensive L2 exposure that are in need of attention. First, we need to know the effects of short-term stays abroad, which are currently underresearched despite their increasing popularity. Second, there is a need for more information on the potential benefits for advanced L2 learners, who thus far do not figure prominently in the existing research. Third, the development of L2 skills beyond oral fluency needs to be further elucidated, not least because the small amount of current research is inconclusive. Finally, individual differences, in particular cognitive capacity, must be considered. Considerations of cognitive capacity in SA research will not only inform potential differences in its effects in these contexts as compared to the robust effects in existing L2 classroom research, but also add to

our knowledge of the relationship between cognition and SA experience.

The current study seeks to address these gaps in the following ways. First, we test L2 learners who are participating in an intensive short-term stay abroad. Second, we test learners with advanced target language proficiency. Third, we focus on the development of morphosyntactic and lexical accuracy and speed. Finally, because conducting a between-groups study on AH (formal experience) versus SA (immersion experience) learners may introduce a number of confounding extraneous variables (e.g., self-selection bias, differences in motivation and attitude, risk-taking, amount and quality of input in two environments) and prohibits random distribution (Sanz, 2014), we conduct a within-subjects study focused exclusively on the effects on language development of intensive L2 exposure in a study abroad context. This methodological aspect is especially important considering that our main goal is not to investigate the superiority of one type of context over another but, rather, to see what short-term study abroad can do and how variation in learner-internal IDs may factor into the effects of such an experience.

In the following section, we review previous studies that have investigated grammatical or lexical development in L2 Spanish during study abroad and the role of cognitive IDs in such development.

BACKGROUND

Individual Differences in Cognition and L2 Development in Study Abroad Contexts

Gaining L2 proficiency as an adult is a complex cognitive action that is likely constrained by a limited system of domain-general cognitive resources (Sagarra, 2013). In particular, variation in one's ability to temporarily store and process incoming information (i.e., working memory capacity, WMC; Baddeley, 2007, 2010) and the ability to phonologically code and store sounds (i.e., phonological working memory, PWM; Baddeley, 1986, 2003) may account for the differential success in L2 learning observed in various exposure contexts. Of interest to the current study is research that has investigated the relationship between the development of L2 Spanish and cognitive variables in an intensive immersion context abroad, namely, processing speed (Segalowitz & Freed, 2004), PWM (O'Brien et al., 2007), WMC (Sunderman & Kroll, 2009; Tokowicz et al., 2004), and attentional or

inhibitory control (LaBrozzi, 2012). Relevant details of each study are summarized in the following section.

Cognitive IDs and Oral Fluency

One of the first studies to investigate the role of cognitive differences during study abroad was by Segalowitz and Freed (2004), who tested two groups of L2 learners on oral performance: one at the beginning and end of a semester in a university L2 Spanish classroom ($n=18$; 13-week semester) and one at the end of their SA exposure in Spain ($n=22$; 13 weeks of SA). Both groups were English L1–Spanish L2 learners with at least two previous semesters of formal study and no previous SA experience. Eight measures of oral fluency (including indicators of temporal and hesitant phenomena) were gathered via an Oral Proficiency Interview (OPI) at the beginning and end of the exposure period. Cognitive processing ability was measured in terms of speed (reaction time, RT) and efficiency (coefficient of variation, CV) in lexical access and attentional control in both the L1 and the L2; L2-specific processing speed and efficiency was indexed by partialling out L1 from L2 RTs and CVs. Results revealed that SA learners made significantly more gains in oral performance than AH learners, regardless of total reported out-of-class L2 contact and previous years studying Spanish, and overall gains in oral proficiency were significantly related to faster and more efficient L2-specific cognitive processing skills, regardless of the learning context.

However, the relationship identified between rate of gain in L2 oral performance and cognitive ability measured in the L2 could stem from a confound with L2 proficiency (Harrington & Sawyer, 1992; Linck et al., 2013; Van den Noort, Bosch, & Hudgahl, 2006). In other words, when WM tasks are administered in the L2, any relationships found likely index both WM abilities and learners' L2 proficiency, rather than domain-general cognitive ability. Though L1 RTs and CVs were used as a baseline measure, they did not report a relationship between processing speed measured in the L1 and efficiency and rate of oral development in L2 Spanish, which would strengthen the interpretation of the reported association. Overall, as recommended by Linck et al. (2013), Van den Noort et al. (2006), and others, cognitive ability should be assessed in the L1 rather than the L2, as done in the next study reported.

As part of a series of studies conducted by Segalowitz and colleagues (Collentine, 2004; O'Brien et al., 2006), O'Brien et al. (2007) compared the relationship between L1 phonological working memory and L2 oral fluency development in two exposure contexts: a regular university classroom setting (AH group) and a study-abroad setting that involved both formal classroom learning and exposure to the native speech community in out-of-class language contact (SA group). Forty-three English L1–Spanish L2 learners completed a serial nonword recognition (SNWR) task in English to measure PWM and a Spanish OPI interview at the beginning and end of the semester. Results showed that both SA and AH learners improved their oral fluency at the end of the semester, but the SA group showed a superior rate of improvement over time. Furthermore, after calculating a residualized change score (Time 1 – Time 2) on each Spanish fluency variable and partialling out learning context effects, PWM was significantly associated with L2 oral fluency development.

To summarize, both Segalowitz and Freed (2004) and O'Brien et al. (2007) found that a SA context promotes a superior rate of improvement in oral fluency relative to the AH setting over the course of a semester for low proficiency learners, and this development was significantly related to variation in L2-specific processing speed and efficiency (Segalowitz & Freed, 2004) and L1 phonological working memory (O'Brien et al., 2007) regardless of context. Nonetheless, further empirical evidence is needed to determine the extent to which aspects of cognitive capacity mediate development in an intensive language exposure context for higher proficiency learners, and using cognitive measures administered in the learners' native language, as done in O'Brien et al. and the following studies.

Cognitive IDs and Lexical Development

Sunderman and Kroll (2009) investigated the role of L1 WM resources in lexical comprehension and production based on previous study abroad experience (SAE). Forty-eight English L1–Spanish L2 adult learners with an average of 4.7 previous semesters of Spanish were grouped according to no SAE ($n=14$) or 2–10 months SAE ($n=34$). They completed translation recognition and picture-naming tasks to measure speed and accuracy of lexical comprehension and production, respectively. WMC was measured in

an L1 Reading Span task (Waters & Caplan, 1996) and analyzed as a continuous variable in order to identify a possible threshold of internal cognitive resources. Results showed that SAE learners were descriptively, though not significantly, faster and more accurate in comprehending and producing words in L2 Spanish than were learners without SAE. Furthermore, a marginally significant interaction between SAE and lexical production led authors to claim that internal resources may mediate the benefits of SA experience and that individuals who lack a certain threshold of WMC are less likely to benefit from the SA context in terms of accurate L2 lexical production.

However, a key limitation of this interpretation is that both groups were tested in the L1 setting with approximately 12.8 weeks between the SAE and testing time, roughly equivalent to a semester. This complicates reliably attributing L2 development to a SA environment. There was also a wide range of variation in time spent abroad within the SAE group (2–10 months), not to mention other unreported differences that likely exist between SA programs (e.g., type of course work, student housing, guided cultural interaction, etc.).

Tokowicz et al. (2004) also investigated the impact of L1 WMC, this time using an L1 Operation Span task (Turner & Engle, 1989), and the effects of relative amount of previous SAE (1–15 months) on oral lexical translation ability (L1 to L2). Fifteen English L1–Spanish L2 learners and twenty-two Spanish L1–English L2 learners were grouped in terms of more (≥ 15 months) and less (≤ 8 months) SAE and higher (5–6) or lower (0–4) WM span based on the set size participants were able to recall. Learners were classified as ‘relatively proficient’ based on a composite average self-rating of 7.17 to 9.22 using a scale of 1–10, with 1 indicating the lowest level of ability and 10 indicating the highest level of ability in reading, writing, conversational ability, and speech comprehension. Translation performance was operationalized as the percentage of the total number of errors that were meaning errors (e.g., incorrect responses related in meaning to correct translation) versus non-response errors (e.g., ‘I don’t know’) on a task in which learners read and orally translated L1 words into their L2. Results revealed that performance accuracy did not statistically differ according to SAE and WMC variables, though the authors note that learners with more SAE were descriptively more accurate than those with less SAE, and individuals with higher WMC were more accurate than those with lower WMC. Additionally, the

group with relatively more SAE and higher WMC had higher circumlocution and made equal numbers of meaning and nonresponse errors, which the authors interpreted as evidence that more SAE and more WMC promote circumlocution ability.

However, similar to the methodological limitation in Sunderman and Kroll (2009), these learners varied considerably in terms of previous time spent abroad and SA program characteristics. Also, learners were tested in their respective L1 environments after returning from the SA setting, the English L1 group in the United States and the Spanish L1 group in Mexico. Thus, these methodological issues limit the impact and interpretation of findings.

In sum, studies by Sunderman and Kroll and Tokowicz et al. provide suggestive evidence that working memory resources play a role in L2 lexical development in an SA setting given that learners in both studies were tested in the L1 environment several months *after* returning from abroad. As a result, amount of L2 exposure and instruction after returning are uncontrolled factors, which weakens their findings pertaining to the role of cognitive capacity in the effects of SA on L2 lexical development.

Cognitive IDs and Lexical and Morphological Processing

LaBrozzi (2012) investigated the extent to which classroom learners with a morphologically poor L1 (English) process morphological and lexical cues to assign verbal tense in a morphologically rich L2 (Spanish) after a period of study abroad. He also investigated the degree to which inhibitory control interacts with previous SA experience in mediating the ability to process redundant morphological/lexical cue incongruencies within a sentence. Sixty intermediate English L1–Spanish L2 learners participated in the study; 36 learners had spent 16 weeks in an immersion environment and returned from their respective programs approximately 3 months prior to time of testing, and 24 learners had no SA experience. Speed and accuracy in inhibitory control—defined as the ability to maintain and process information relevant to a task while suppressing extraneous information—were measured via a Simon task, which, across trials, requires inhibition of potentially distracting task-irrelevant information (such as color or location of stimuli) to achieve task goals. To determine which cues learners relied on to resolve adverb–verb tense

incongruencies, eye-tracking measured the length of time learners initially took to read the relevant information (i.e., first pass gaze duration), total cumulative reading time, and the percentage of regressions, or how many times they visually returned to the site of incongruency in the sentence.

Based on these data, LaBrozzi concluded that the SAE group relied more on morphological cues (verbs) than the group without SAE, but nonetheless continued to rely on lexical cues (adverbs) to assign temporal reference. The author attributed this to learners' intermediate, transitional stage in approximating native-like processing patterns in the L2. However, again, learners were tested 3 months after returning from SA, making it difficult to directly relate these results to the SA context of L2 exposure. Inhibitory control was not found to influence processing strategies for either the SAE or nonSAE group (i.e., it did not serve to block attention to L1-based lexical cues and shift reliance to more informative L2 morphological cues).

Motivation for the Current Study

The small number of studies detailed in the previous sections contribute to an increased understanding of relationships between cognitive ability and the development of Spanish L2 oral fluency (O'Brien et al., 2007; Segalowitz & Freed, 2004), lexical comprehension and production (Sunderman & Kroll, 2009), oral circumlocution ability (Tokowicz et al., 2004), and morphological processing strategies (LaBrozzi, 2012) following intensive language exposure abroad. However, the methodological variability and limitations along with the mixed results make it challenging to draw clear conclusions. First, time spent abroad not only varied widely within the SA groups themselves in these studies but also followed the general trend of investigating L2 development over semester or year-long programs rather than short-term programs abroad. This is a considerable limitation given the latest IIE data reporting that nearly 60% of all U.S. students participate in programs lasting 8 weeks or less. In line with arguments made by Llanes (2011) and Llanes and Muñoz (2009), the increasing popularity and economic value of intensive summer and winter study abroad programs demand that we provide more detailed information about the profile of L2 development during shorter periods of immersion.

Second, learners in the reviewed studies were at novice, intermediate, or mixed levels of L2 proficiency, except for Tokowicz et al. who investigated relatively proficient learners based on self-report and Sunderman and Kroll whose learners had 4.7 semesters of previous instruction on average. Overall, the efficacy of SA in more advanced L2 learners as well as the nature of potential relationships between the benefits of immersive language exposure and cognitive ID variables in such learners remains an open question and one that requires empirical attention in light of DeKeyser's (1991, 2007, 2010, 2014) theoretical claim that learners who have more explicit knowledge to draw on and who possess more communicative skills are better poised to automatize their L2 skills compared to less advanced learners.

Third, the measures of L2 development in all studies focused on oral fluency, speed and accuracy of lexical comprehension and production, or processing of morphological and lexical cues, but we still know little about the association between cognitive resources and accuracy or speed in morphosyntactic L2 development under intensive language exposure.

Finally, with the exception of Segalowitz and Freed (2004) and O'Brien et al. (2007), all studies tested participants in the AH/L1 context after variable periods of time following their return from the immersion environment, which introduces uncontrolled factors such as (unreported or unknown) differences between the original SA programs that learners participated in and the amount of L2 exposure and instruction between their return and testing time.

To address some of the gaps and limitations in previous research, the current study investigated the efficacy of short-term intensive language exposure abroad for promoting grammatical and lexical sensitivity to L2 Spanish in advanced L2 learners. Learners were tested at the beginning of their immersion exposure and five weeks later, at the end of the summer program. Learners also completed several memory tasks in order to investigate the extent to which internal cognitive resources may be related to the effects of the external immersion context on L2 development. The program and all testing were completed in Barcelona, Catalonia, Spain.

RESEARCH QUESTIONS

Our study aimed to address the following questions:

- RQ1. Does a 5-week L2 intensive language experience abroad lead to improved L2 grammatical and/or lexical development in advanced learners, as measured by accuracy and latency?
- RQ2. Are working memory and/or phonological working memory related to the degree of L2 grammatical and lexical development in advanced learners over a 5-week intensive language experience abroad?

METHODOLOGY

Participants

The participants in the study were 26 L2 learners of Spanish (mean age = 20 years; 19 female, 7 male). These learners had an average of 6.4 ($SD = 2.31$) years of Spanish study in a classroom setting prior to beginning the study abroad program. In order to qualify for this study abroad program, the learners had to have taken two advanced (3rd year) Spanish language classes prior to applying for the SA program and have a GPA of 2.7 or higher. None of the participants had studied abroad prior to their time in Barcelona and none were fluent in any language aside from English, their L1.

Context of the Intensive Exposure Abroad

The full program consisted of 5 weeks of immersive coursework and fieldwork plus an additional half a week in Barcelona before the beginning of classes. Prior to beginning the program, all of the learners signed a Language Pledge affirming their commitment to communicating only in Spanish during the entire duration of the program; violations were penalized with grade reductions or even suspension from the program. During the program, students were enrolled in three content courses or the equivalent of nine credits (students generally complete 12–15 credits per semester, so this was a highly intensive program), which were taught in Spanish by faculty from both the home institution, Georgetown University, and from the host institution, Universitat Pompeu Fabra, on topics covering visual arts, history, and politics (12 hours/week with an average of 10 hours of homework per week). The students also participated in faculty-led fieldwork (14 hours/week), and had regular meetings with native Spanish-speaking language exchange partners (2–6 hours/week). The home institution's attendance policy and participation grades applied to fieldwork, conversation sessions, and coursework.

The program was designed in 2007 by an applied linguist with several goals in mind: (a) to optimize to the maximum the quality and quantity of L2 oral and written input, making it constant and authentic, (b) to increase interactions among all program participants, widely understood as faculty, graduate assistants, local assistants, and local and program students, thereby increasing the opportunities for practice, and (c) to help develop a network of knowledge of the history and politics of identity of the region and its expression through the visual arts based on a combination of lectures and readings as well as primary sources available during fieldwork, all in the L2. To achieve this, students were always busy with class work (at least 9 brief papers, 6–9 exams, 3–6 presentations) as well as review sessions, conversation exchanges, and fieldwork.² Days started at 10am and finished at 9pm four days a week. An added advantage of keeping students busy in a city like Barcelona is that they were less prone to get in trouble, so by the time data collection began for this project, voluntary activities had been included, which secured even more participation and exposure to the target language. Many details were worked out to ensure constant access to Spanish including (a) the presence of the assistants' room in the residence hall, (b) faculty, assistants, and native speaker guests at the table during meals, (c) having classes in the main building rather than the new building occupied by other study abroad programs, (d) housing students in the residence hall away from other summer programs; and (e) prohibiting cell phone use during group activities.

Materials

Grammaticality Judgment Task. The grammaticality judgment task (GJT) was designed using SuperLab 4.0 (Cedrus Corp, San Pedro, CA) and presented in the visual modality. Prior to beginning the task, participants were provided with written instructions in English, which were also read aloud by the experimenter. Participants were informed that they would be reading sentences in Spanish and that their task was to decide as quickly and accurately as possible whether each sentence was grammatical or not. To ensure that participants understood the goal of this judgment task, they were provided with one example each of a grammatical and ungrammatical Spanish sentence. During the task, full sentences were displayed one at a time in gray 20-point font in the center of a black screen. Sentences were

4–10 words in length ($M=5.60$, $SD=2.33$) and each one was visible on the screen until the participant responded via button press or was timed-out after 4000 milliseconds (ms), after which the next sentence automatically appeared. There were five practice sentences prior to the experimental items. Both accuracy and reaction time were recorded; participants were not provided with feedback during the task. Items were presented randomly for each participant.

Two versions of the GJT were created and each consisted of 60 items selected from the materials used in Bowden et al. (2013). The linguistic targets in the present study were word order, noun–adjective number agreement, and noun–adjective gender agreement. A sample of the experimental items can be found in Table 1. In each version of the GJT, there were 30 grammatical and 30 ungrammatical items. Of the 30 ungrammatical items, 10 contained an error in word order, 10 contained an error in noun–adjective number agreement, and 10 contained an error in noun–adjective gender agreement. The difference between the two versions was such that an ungrammatical sentence in Version A was a grammatical sentence in Version B. These versions were counter-balanced within and across participants.

Lexical Decision Task. The lexical decision task (LDT) was also presented in the visual modality³ (Superlab 4.0; Cedrus Corp, San Pedro, CA) and participants were provided with written instructions in English that were also read aloud by the experimenter prior to beginning the task. They were informed that they would be presented with a letter string and that their task was to decide as quickly and accurately as possible whether the string was a word in Spanish or not. Again, to ensure they understood the goal of their decision, they were provided with one example of a word and one example of a nonword. Each trial began with a 500 ms fixation (crosshair) followed by the letter string, presented in gray 20-point font in the center of a black screen. Each letter string

remained visible until the participant responded via button press or was timed-out after 1500 ms, after which the next trial began. There were five practice items prior to the experimental items and all experimental items were presented randomly for each participant throughout the task. Both accuracy and reaction time were recorded; participants were not provided with feedback on their responses.

There was a single version of the LDT that consisted of 160 items. A single version was used to ensure comparability of measuring lexical processing of the same experimental items over time, in order to determine whether performance within these lexical items changed as a function of the learners' study abroad experience, as it is not possible to create comparable but different versions of a lexical knowledge/processing test (for administration of the same task versions as pre- and post-test measures in other SA/lexical processing research see Akamatsu, 2008; Llanes & Muñoz, 2009; Segalowitz & Freed, 2004; among others). Of the 160 items, 80 were words (noncognates; 40 concrete Spanish nouns, like *ventana*, 'window,' and 40 abstract nouns, like *verdad*, 'truth') and 80 were phonologically and orthographically legal nonwords.

The 80 Spanish nouns were confirmed to be present in Spanish language textbooks used at the U.S. university where the students took formal Spanish classes prior to the SA program (Dominicis & Reynolds, 2011; VanPatten, Leaser, & Keating, 2007; VanPatten et al., 2008, 2010). They were normed for frequency using the online database Corpus del Español (Davies, 2002). There was no difference in the frequency⁴ of abstract and concrete nouns $t(78) = 1.66$, $p = .100$ (mean abstract frequency = .00277, mean concrete frequency = .00203). There was also no difference in the word length of abstract ($M = 5.92$ letters) and concrete ($M = 5.75$ letters) nouns, $t(78) = 0.536$, $p = .593$. The 80 nonwords were created by replacing a single letter of a noun used in the word list while maintaining

TABLE 1
Sample Grammatical and Corresponding Ungrammatical Items on the Grammaticality Judgment Task

	Grammatical	Ungrammatical
Word order	Tengo que correr muchas millas esta semana	Tengo que !millas muchas correr esta semana
Number agreement	La falda es nueva	La falda es !nuevas
Gender agreement	El vestido es negro	El vestido es !negra

Note. ! = violation of the linguistic target.

phonological and orthographic acceptability (e.g., *ventapa*, *gerdad*).

Sentence Span Working Memory Task. A modified version of Daneman and Carpenter's (1980) sentence span task was used in this study (see also Lado, 2008; Sanz et al., 2015). During the task, participants listened to a series of 48 sentences (in English) that were presented in sets of three, four, or five sentences. For each sentence, participants were asked to judge whether the sentence made sense and whether it was grammatical. The ungrammatical sentences contained errors in subject–verb agreement, verb tense, or an article, and nonsense sentences contained animacy violations. After each set, participants heard a tone that signaled them to recall the final word of each sentence in that set. The sentences were presented auditorily; participants could not re-listen to any items and were not provided with feedback on their judgments. Participants made their grammar/sense judgments on paper and their recall was audio-recorded; both grammar/sense and recall performance were later coded for accuracy. Accuracy was calculated by assigning 1 point for a correct grammatical judgment, 1 point for a correct sense judgment, and 1 point for each correctly recalled word in a set, for a total of 144 maximum possible points.

Phonological Working Memory Tasks. Two non-word repetition tasks (NWRTs) were used to assess the participants' L1 and L2 phonological working memory capacity. The two tasks were identical with the exception of the language they were designed to approximate. In the L1 NWRT, the nonwords obeyed English phonotactics (*melistok*, *nutolon*) while in the L2 NWRT, the nonwords obeyed Spanish phonotactics (*asaco*, *filomo*). Each task consisted of sixteen 2-item sets of nonwords. From set 1 to set 16, the number of syllables-per-item increased from 3 to 8, requiring participants to hold 16 syllables in memory by the last set. Participants were instructed to listen to each set and then repeat the words aloud after they heard a tone. Participants' verbal repetitions were audiorecorded and later coded for accuracy. Both NWRTs were scored by assigning 1 point for each correctly repeated word in a set without more than one mistake over the 2-item set; the maximum possible score on each of these tasks was 32. The items for the L1 NWRT were prerecorded by a female native speaker of American English (Lado, 2008). The items for the L2 NWRT were prerecorded by a male native speaker of Spanish (Grey, Williams, & Rebuschat, 2015).

Procedure. After providing written informed consent to participate in the study, participants were scheduled for individual sessions with a researcher. Each participant met with a researcher one time every week during the 5-week program. During Week 1 and Week 5 the GJT and LDT were administered. Participants completed the sentence span working memory task in Week 2, the L1 NWRT in Week 3, and the L2 NWRT in Week 4. Other tasks, including oral and written production, survey instruments, and additional cognitive tasks were also completed during each of the five sessions but are not discussed here as they remain outside the scope of this paper.

RESULTS

The data were analyzed using SPSS (version 21.0, IBM Corp.). Percent accuracy and mean reaction times were entered into analyses for both the GJT and LDT. Only reaction times to correct responses were considered; error-related RTs were replaced with the item mean (Lachaud & Renault, 2011). To measure potential changes in grammatical sensitivity from Week 1 to Week 5, two-tailed paired-samples *t*-tests compared Week 1 and Week 5 accuracy and latency on the GJT. A similar set of *t*-tests was done comparing Week 1 and Week 5 accuracy and latency for the LDT (α set at .05 for both tasks). In reporting our results we include effect sizes and follow the guidelines suggested by Oswald and Plonsky (2010) for interpreting Cohen's *d* in SLA research: $d = .40$ is a small effect size, $d = .70$ is a medium effect size, and $d = 1.00$ is a large effect size. To measure whether any of the three memory tasks were related to changes in performance on the GJT or LDT, gain scores were calculated. For accuracy, this was done by subtracting mean accuracy at Week 1 from mean accuracy at Week 5 for each participant. The same was done for latency; here, Week 5 RTs were subtracted from Week 1 RTs, as gains would be signaled by reductions in latency over time and reflected in a negative value. Then, all mean change scores were entered into correlation analyses with the cognitive capacity measures. For the correlations, we used a Bonferroni-adjusted alpha of .005 to correct for multiple comparisons.

Grammaticality Judgment Task

Means, standard deviations, and standard error for accuracy and reaction time on the GJT at Week 1 and Week 5 are reported in Tables 2 and 3 for the three targets of word order, number

TABLE 2
Descriptive Information on Accuracy Performance at Week 1 and Week 5 for the Grammaticality Judgment Task

	Week 1			Week 5			Δ Accuracy
	<i>M</i>	<i>SD</i>	<i>SE</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	
Total*	55.12	23.37	4.58	67.94	14.62	2.86	12.82
Word Order*	49.03	23.41	4.59	61.92	17.26	3.38	12.88
Number*	67.30	32.77	6.42	88.65	18.63	3.65	21.34
Gender	70.38	27.85	5.46	83.46	19.06	3.73	13.07

Note. Number and Gender refer to nominal agreement; *M*=mean, *SD*=standard deviation, *SE*=standard error. * = significant difference between Week 1 and Week 5 at $p < .05$. Δ Accuracy = mean accuracy change used for correlation analyses.

agreement, and gender agreement. Overall, learners significantly improved in their accuracy in judging items after 5 weeks in an immersion environment, $t(25) = -2.56$, $p = .017$, $d = .50$. These learners also showed a significant decrease in RTs after 5 weeks, $t(25) = 4.00$, $p < .001$, $d = .78$. When considering the three linguistic targets of word order, number agreement, and gender agreement separately, a more nuanced pattern of results emerged. For both word order and number agreement, accuracy in judging items showed a significant improvement from Week 1 to Week 5, $t(25) = -3.32$, $p = .003$, $d = .65$ and $t(25) = -3.14$, $p = .004$, $d = .61$, respectively (Figure 1). For gender agreement, the difference in accuracy only approached significance $t(25) = -1.95$, $p = .062$, $d = .38$. Finally, from Week 1 to Week 5 there was no significant difference in RTs for any of the three targets analyzed separately, all $ps > .05$, all $ds < .25$.

Lexical Decision Task

Means, standard deviations, and standard error for accuracy and RT on the LDT at Week 1 and Week 5 are reported in Tables 4

and 5. Overall, there was significant improvement in accuracy and a significant decrease in RTs after 5 weeks in an immersion context, $t(25) = -3.97$, $p = .001$, $d = .77$ and $t(25) = 3.58$, $p = .001$, $d = .70$ (Figure 2). Performance on words and nonwords were also analyzed separately. For words, there was no change in accuracy after 5 weeks, $t(25) = -1.59$, $p = .123$, $d = .31$, but there was a significant decrease in RT, $t(25) = 2.43$, $p = .023$, $d = .47$. For nonwords, learners became both significantly more accurate in identifying these letter strings as nonwords in Spanish, $t(25) = -5.28$, $p < .001$, $d = 1.03$ and faster, as evidenced by a significant difference in RTs from Week 1 to Week 5, $t(25) = 3.93$, $p = .001$, $d = .77$.

Gains And Memory Tasks

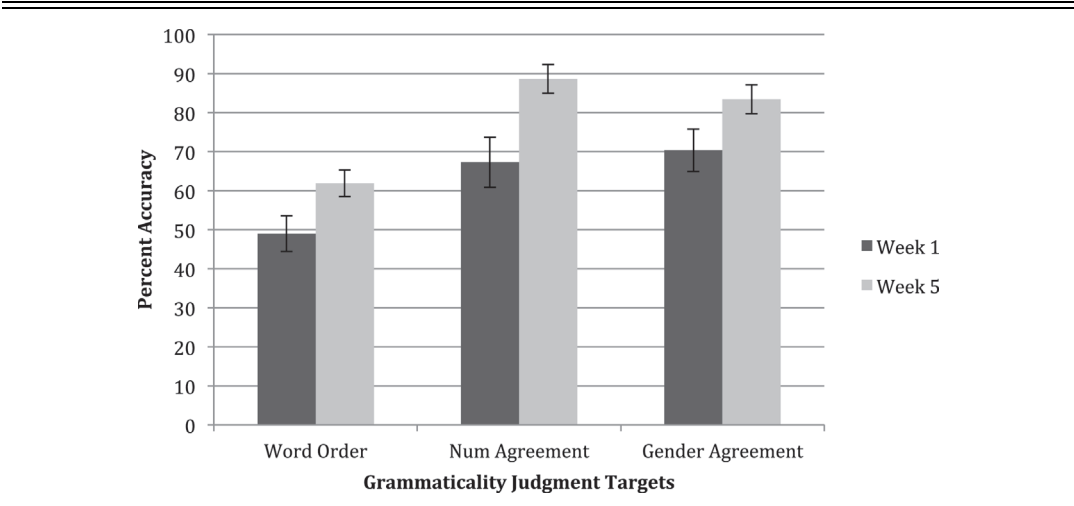
In order to determine whether cognitive capacity was related to L2 development over the study abroad period, difference scores were calculated for performance on the GJTs and LDTs (mean change scores can be found in Tables 3, 4, and 5; for performance on the measures of cognitive capacity, see Table 6). The coefficients

TABLE 3
Descriptive Information on Reaction Time Performance at Week 1 and Week 5 for the Grammaticality Judgment Task

	Week 1			Week 5			Δ Latency
	<i>M</i>	<i>SD</i>	<i>SE</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	
Total*	2231.3	332.6	65.2	1991.1	219.4	43.0	-240.18
Word Order	2715.2	769.9	150.9	1720.0	352.8	69.2	-12.73
Number	1768.6	653.1	130.6	1720.0	249.9	49.9	-48.59
Gender	1875.7	387.3	75.9	1784.9	266.9	52.3	-90.76

Note. Number and Gender refer to nominal agreement; *M*=mean, *SD*=standard deviation, *SE*=standard error; * = significant difference between Week 1 and Week 5 at $p < .05$; values are reported in milliseconds; Δ Latency = mean latency change used for correlation analyses.

FIGURE 1
Accuracy Performance on the Separate Linguistic Targets for the Grammaticality Judgment Task



Note. Num = Number.

TABLE 4
Descriptive Information on Accuracy Performance at Week 1 and Week 5 for the Lexical Decision Task

	Week 1			Week 5			ΔAccuracy
	<i>M</i>	<i>SD</i>	<i>SE</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	
Total*	79.81	11.5	2.25	85.72	7.88	1.54	6.53
Word	81.63	19.03	3.73	86.49	8.28	1.62	4.85
Nonword*	76.73	15.42	3.02	84.95	14.48	2.84	8.22

Note. *M* = mean, *SD* = standard deviation, *SE* = standard error; * = significant difference between Week 1 and Week 5 at $p < .05$; ΔAccuracy = mean accuracy change used for correlation analyses.

TABLE 5
Descriptive Information on Reaction Time Performance at Week 1 and Week 5 for the Lexical Decision Task

	Week 1			Week 5			ΔLatency
	<i>M</i>	<i>SD</i>	<i>SE</i>	<i>M</i>	<i>SD</i>	<i>SE</i>	
Total*	820.3	89.0	17.4	771.6	78.0	15.4	−48.65
Word*	764.2	73.4	14.3	730.8	60.9	11.9	−33.33
Nonword*	876.4	116.9	22.9	812.4	105.4	20.6	−63.97

Note. *M* = mean, *SD* = standard deviation, *SE* = standard error; * = significant difference between Week 1 and Week 5 at $p < .05$; values are reported in milliseconds; ΔLatency = mean latency change used for correlation analyses.

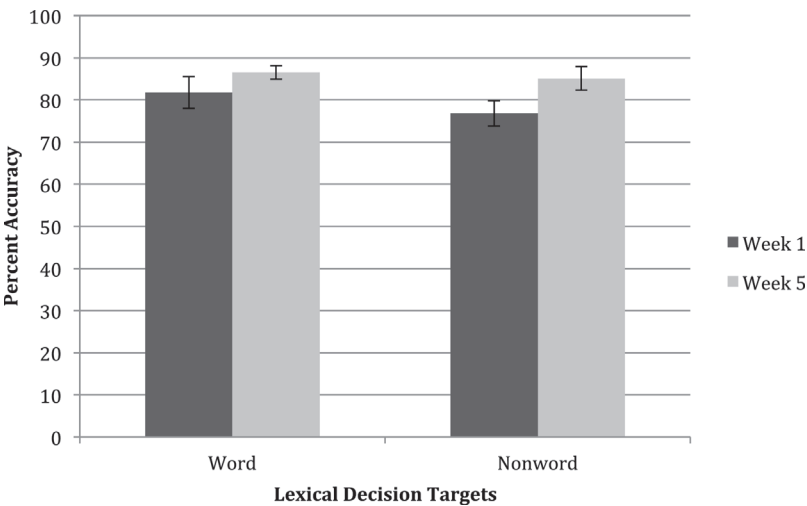
for correlations between GJT and LDT accuracy difference scores and the cognitive capacity measures (sentence span L1 WM, L1 NWRT, L2 NWRT⁵) can be found in Table 7. No significant correlations were found. Table 8 reports the coefficients for correlations between GJT and LDT latency reductions and the three measures of cognitive capacity. From this, one significant positive correlation returned, between L1 NWRT

and overall GJT latency reduction, $r(26) = .573$, $p = .002$.

DISCUSSION

The primary goal of this study was to investigate the extent of advanced L2 learners' grammatical and lexical development in the context of short-term intensive L2 exposure abroad in order to

FIGURE 2
Accuracy Performance on Word and Nonword Judgments in the Lexical Decision Task



elucidate conflicting findings in both linguistic domains, where some studies find gains (e.g., Dewey, 2008; Duperron, 2006; Ife et al., 2000; Isabelli & Nishida, 2005; Juan-Garau et al., 2014) and some do not (Collentine, 2004; DeKeyser, 1991, 2010; Isabelli-Garcia, 2010). A short-term period abroad was considered ideal due to the increasing popularity of such programs with U.S. students (IIE, 2013). In addition, selection of advanced-proficiency learners has both practical and theoretical importance (DeKeyser, 1991, 2007, 2010, 2014) and yet few such studies exist (Ortega & Byrnes, 2008).

A secondary goal of the study was to investigate whether cognitive capacity would be related to any within-learner linguistic gains over the 5-week intensive language experience abroad. This goal was achieved by examining the relationship

between gains in L2 grammar and vocabulary with measures of cognitive capacity, specifically, WM and PWM. Such relationships are relatively well documented in nonintensive classroom contexts (Juffs & Harrington, 2011; Linck et al., 2013) but the extent of their presence is less clear in a SA setting (LaBrozzi, 2012; O'Brien et al., 2007; Segalowitz & Freed, 2004; Sunderman & Kroll, 2009; Tokowicz et al., 2004).

In addition, methodological improvements were made to previous studies. We maximized the homogeneity of the sample by administering computerized tests *in situ* and also by having strict

TABLE 6
Descriptive Information on Cognitive Capacity Measures

	M	SD	SE
Sentence Span	104.0	22.7	4.4
L1 NWRT	19.6	.74	3.8
L2 NWRT	12.8	.62	3.1

Note. M = mean, SD = standard deviation, SE = standard error; L1 NWRT = English nonword repetition task, L2 NWRT = Spanish nonword repetition task. Maximum possible score on each NWRT was 32, and 144 for sentence span.

TABLE 7
Intercorrelations Between Grammaticality Judgment and Lexical Decision Task Accuracy Change and Measures of Memory Capacity

	Sentence Span	L1 NWRT	L2 NWRT
GJT overall	.082	-.229	-.081
Word order	-.042	-.074	-.014
Number agr.	.130	-.220	-.032
Gender agr.	.033	-.262	-.135
LDT overall	.086	-.037	-.124
Word	.168	.139	.121
Nonword	.169	.035	-.076

Note. Value reported is Pearson correlation coefficient. L1 NWRT = English nonword repetition task, L2 NWRT = Spanish nonword repetition task, agr. = agreement.

TABLE 8
Intercorrelations Between Grammaticality Judgment
and Lexical Decision Task Latency Change and
Measures of Memory Capacity

	Sentence Span	L1 NWRT	L2 NWRT
GJT overall	-.200	.573*	.020
Word order	-.091	.225	.210
Number agr.	-.031	-.109	-.346
Gender agr.	.143	.281	.083
LDT overall	.115	.179	-.180
Word	.202	.008	-.167
Nonword	.021	.291	-.159

Note. Value reported is Pearson correlation coefficient. L1 NWRT = English nonword repetition task, L2 NWRT = Spanish nonword repetition task, agr. = agreement. Latency values are reported in milliseconds; * = significant correlation at $p < .005$ (Bonferroni-corrected).

linguistic and academic entrance requirements. Computerized tests allow for collecting latency measurements, a finer-grained measure of change in L2 processing, in addition to traditional accuracy measurements (see Sanz & Grey, 2015). In our case, without latency data, we would not have seen any changes in processing of L2 words since participants' accuracy of accepting words did not change over time. Additionally, the LDT is an improvement over traditional vocabulary tasks, as it can easily test a large number of words and reduces the potential for affecting outcomes due to test-taking strategies (Kempe & MacWhinney, 1996). Administering such tests *in situ* removed the potentially confounding factors of presence (or lack of) post-SA L2 exposure and focused the scope of the study on real-time effects of the intensive L2 exposure. Lastly, a homogeneous *in situ* sample avoided potential confounds in a between-subjects (AH versus SA) design, such as self-selection bias, proficiency level, living situation, and previous SA experience.

We next discuss the main findings of the study and conclusions from our research.

Main Findings

This study investigated L2 development in accuracy and speed of judging L2 morphosyntactic patterns (on the GJT) and L2 lexical items (on the LDT). Participants significantly improved in their accuracy and speed on both tests over the course of the intensive exposure period. Therefore, we conclude that advanced profi-

ciency learners can and do improve their L2 abilities during a 5-week intensive experience abroad. This finding supports DeKeyser's (2007, 2010, 2014) proposal that advanced proficiency learners are well poised to significantly benefit from study abroad and corroborates previous empirical findings (e.g., Duperron, 2006; Isabelli & Nishida, 2005, for morphosyntax).

When the three grammatical targets tested on the GJT were analyzed separately, our results show that significant improvements to learners' accuracy were limited to judgments of word order and number agreement; there were no significant changes in accuracy of gender agreement judgments. The lack of significant improvement for gender agreement is not surprising, given that this feature is notoriously difficult for late L2 learners whose L1 does not have grammatical gender, both in traditional classroom settings and in more immersion-like conditions (e.g., Alarcón, 2009; Cambor-Portilla, 2006; Grey, 2013; Morgan-Short et al., 2010, for L2 Romance languages; Davidson & Indefrey, 2009, for L2 Germanic languages). Additionally, latency did not change significantly over the course of the 5 weeks for the three targets separately. This is likely due to the loss in power caused by reducing the number of items (10 for each target separately compared to 30 combined items), and thus does not provide sufficient evidence to say that learners did or did not improve their speed in responding to each target structure.

When examining the lexical results from the LDT in detail it appears that, over the period of intensive exposure abroad, learners became more accurate in correctly identifying nonwords as such, whereas their accuracy in correctly identifying Spanish words did not change significantly. Their speed in responding, however, significantly decreased both when the stimuli were words and when they were nonwords. Although accuracy was descriptively higher for words than nonwords at Week 1, which could potentially create more room for improvement in nonwords, both words and nonwords had correct response rates below 90% (below ceiling) at Week 1, so it is unlikely that ceiling effects explain these outcomes. It seems, therefore, that latency was affected equally for both nonwords and words and that the intensive exposure abroad had more of an effect on correct rejection of nonwords than on correct acceptance of words.

Under Jiang's (2000) breakdown of lexical performance, our learners' improved speed in correctly accepting words shows an improvement in lexical competence (automatic access to

mental lexical entry) but not lexical knowledge (form, meaning, and usage of the lexicon) or lexical transfer (using L2 words as if they were L1 words; we did not test these aspects). It is more difficult to explain the processes underlying accurate and quick rejection of nonwords. One possible explanation is that the extensive input from the program raised participants' familiarity with L2 lexical items. Following Balota and Chumbley's (1984) familiarity/meaningfulness continuum, increasing familiarity with L2 words would also increase their meaningfulness. In turn, the familiarity and meaningfulness of nonwords would be reduced, thereby making the response times to both types of stimuli faster. Seidenberg and McClelland's (1989) interactive activation framework differs in details of sublexical processes, but also relies on a threshold of familiarity for making yes/no lexical decisions, thereby supporting the explanation of increased familiarity with the lexical items as accounting for the L2 lexical development found in the current study.

Overall, then, our findings for linguistic gains show that advanced-proficiency learners improved their grammatical and lexical accuracy and speed in L2 receptive processing during a 5-week period. Our next area of interest was to identify who amongst the participants might be most likely to experience these gains. We pursued this by testing for relationships between linguistic gains and measures of cognitive capacity, namely WM and PWM capacities.

Correlations revealed no significant relationships between either domain (morphosyntactic or lexical) for gains in L2 accuracy and any measure of cognitive capacity (L1 WM, L1 PWM, or L2 PWM). Correlations between changes in L2 speed of processing and cognitive capacity returned significant only for the relationship between overall change in latency on the overall GJT and L1 PWM; all other relationships were nonsignificant. Thus, advanced-level participants improved in their L2 morphosyntactic and lexical judgment accuracy over the 5 weeks abroad, regardless of potential variation in their cognitive capacity. This contrasts the relatively strong evidence of the role of cognitive capacity found in L2 classroom settings (e.g., Juffs & Harrington, 2011; Linck et al., 2013), in which individuals with higher cognitive capacities generally show more rapid L2 development. It also contrasts previous evidence of the role of cognitive capacity in an SA context (O'Brien et al., 2007; Sunderman & Kroll, 2009; Tokowicz et al., 2004).

The significant correlation between reduced speed when judging grammaticality and PWM

capacity was positive, meaning higher capacities of PWM were associated with less reduction in latency from the beginning to end of the program. This result is unexpected, since previous studies suggest that higher levels of PWM are associated with increased rate of L2 grammar learning (e.g., Martin & Ellis, 2012; Williams & Lovatt, 2003). However, to our knowledge, no other studies to date have looked at the relationship between latency changes and cognitive capacity. Therefore, further research is needed to decide whether this finding is meaningful or spurious. Since nearly all the correlations (for both accuracy and latency) were nonsignificant at a conservative alpha-level of .005, suggesting little or no relationship between cognitive capacity and L2 development in our sample, we did not conduct regression analyses to investigate whether high or low cognitive capacity predicted gains in L2 grammar and lexicon.

Study Characteristics

Placing our findings in the body of SA research requires a closer look at three key aspects of our research design: the short length of the study abroad program (5 weeks), participants' proficiency level (advanced), and the consideration of aspects of cognitive capacity (WM and PWM) in relation to linguistic gains. In the following section, each of these aspects will be discussed in light of our findings.

Program Length. While most SA studies have focused on whether learners show significant gains in L2 development in semester-long and year-long study abroad programs, we took a different approach and instead asked whether learners can benefit from a short-term program abroad. Our affirmative answer to this question adds to evidence of gains in short-term programs reported by other researchers for other L2 skills (Llanes & Muñoz, 2009, for listening comprehension, oral fluency, and oral accuracy; Cubillos et al., 2008, for listening comprehension). It is reassuring for participants and administrators that the observed trend of study abroad experiences taking place for shorter periods of time (whether they are summer programs, January terms, or other programs less than 13 weeks long; IIE Report, 2013) does not appear to preclude the achievement of linguistic gains that for many students are the goal of the study abroad experience. Quality and quantity of input are of primary concern for maximizing L2 gains, and even shorter programs can have the intensive

content-based courses, structured cultural activities, and dedication to use of the L2 evidenced by a language pledge that can contribute to L2 input that is of high quality and quantity.

Proficiency Level. As DeKeyser (2007, 2010, 2014) predicts for advanced learners in SA, our participants seem to have had the linguistic groundwork in place to be able to “hit the ground running” (DeKeyser, 2007, p. 314) upon arrival at the program site, incorporating multiple features of the linguistically complex input around them to yield significant development in L2 morphosyntax and lexicon in just 5 weeks. Two other studies of advanced proficiency learners have also found significant gains for morphosyntax (Isabelli & Nishida, 2005; Juan-Garau et al., 2014). In the case of lexicon, two studies have found gains with advanced (along with intermediate) participants (Dewey, 2008; Ife et al., 2000). For both morphosyntax and lexicon, we extend these findings to gains in a shorter program.

What may have caused these gains in L2 development? Although our study did not isolate the roles of different aspects of SA (e.g., home-stays, number of credit hours), we can speculate as to the causes based on the type of changes seen in learners, particularly in latency. Reductions in latency signify quicker processing of grammar and vocabulary at the end of the program, a feat that seems best explained by the intensive nature of the L2 exposure abroad. As detailed in the Methodology section, input came in the form of intensive content courses taught in the L2 and organized extracurricular activities, such as fieldwork and conversation partners. In the former especially, both aural and written input likely contained robust exemplars of grammatical structures and lexical items. In the case of oral interactions, native speakers required real-time responses from learners. In the case of reading, the courses required large amounts of daily reading of authentic material in the L2. Both types of input and resultant practice may have promoted automatization (DeKeyser, 2014), which in turn could contribute to the quicker processing speed found here. This reduction in speed parallels findings for improved oral fluency as a result of SA (Collentine & Freed, 2004; Freed, 1995; Freed et al., 2004; Llanes & Muñoz, 2009; Llanes et al., 2011; Trenchs-Parera, 2009), as both are related to speed of processing.

Cognitive Capacity. Length of program and proficiency level may each play a role in the lack of significant relationships found between linguistic

gains and any of the three aspects of cognitive capacity measured here (L1 WM, L1 PWM, and L2 PWM). In addition, it is informative to consider the relationship between cognitive capacity and each of the two linguistic domains (grammar and lexicon) separately.

To begin with, our focus on short-term study abroad makes a unique contribution to the field, as the small number of previous studies of cognitive capacity and SA all investigated longer stays abroad. As an extreme example, Tokowicz et al. (2004) classified learners with stays of 8 months or fewer as having a small amount of study abroad experience, though it is much more than our participants' 5 weeks. They found no interaction between length of stay abroad and WM capacity, nor did LaBrozzi (2012) find an interaction between inhibitory control and native-like processing in a semester-long program. These two findings contrast with other studies that have found gains particular to high cognitive capacity individuals in SA settings (e.g., O'Brien et al., 2007, for PWM in a semester-long program; Segalowitz & Freed, 2004, for attentional control in a semester-long program; Sunderman & Kroll, 2009, for the production task only and WM with stays abroad ranging from 2–10 months). Clearly, further studies are needed to confirm our finding that WM and PWM are not related to linguistic gains in short-term programs, and to elaborate on whether the result is a feature of general SA-related gains assessed *in situ* or would differ with other program lengths.

Second, L2 proficiency level may change the relationship between cognitive capacity and L2 learning. Specifically, higher cognitive capacity may be more important at lower levels of proficiency than at more advanced levels of proficiency, with some evidence to this effect in traditional classroom settings (e.g., Cheung, 1996; Hummel, 2009; Fehrer & Fry, 2007a, 2007b; Serafini, 2013). Nonetheless, in a recent meta-analysis consisting of primarily classroom studies, Linck et al. (2013) argue that research remains inconclusive in this area and more studies are needed that contrast the contributions of WM and PWM for higher and lower proficiency learners. In SA research, O'Brien et al. (2007) found a significant positive relationship between PWM and oral fluency, but learners' proficiency level is only vaguely stated (a minimum of two semesters' prior study; no maximum, average, or range given). Sunderman and Kroll (2009) and Tokowicz et al. (2004) provide even less information regarding their participants' proficiency level. Therefore, from our study we can conclude

that advanced-level learners' L2 development during an intensive language experience abroad is not significantly related to their cognitive capacity. Further research is needed to determine whether this result is specific to advanced-level intensive L2 exposure programs or is a general feature of L2 learning during intensive exposure abroad, regardless of proficiency level.

Finally, it is noteworthy that we found no relationship between cognitive capacity and accuracy gains in either of the two L2 domains tested. For grammar, this finding augments LaBrozzi's (2012) finding of no such relationship in morphosyntactic processing at the intermediate level by extending it to advanced learners who participated in a short-term program. It remains to be seen whether changes in speed of processing L2 morphosyntax are related to cognitive capacity or not. Our results suggest a relationship with PWM, but in an unexpected direction. For lexical development, our finding contrasts somewhat with previous findings of a significant relationship between PWM and oral fluency gains during SA (O'Brien et al., 2007) and an interaction between WM capacity and study abroad experience that approached significance (Tokowicz et al., 2004). Notably, the interaction in the latter study was found only on the task of productive lexical processing and not on the receptive lexical processing task. Since our task was also receptive, it may be that receptive lexical processing is less bound by cognitive capacity than productive processing is. Due to the limited number of SA studies that have investigated this issue and the multiple methodological differences between them, our findings on the relationship of linguistic development to cognitive capacity must be replicated in similar studies employing *in situ*, within-subjects, longitudinal designs to directly compare the role that cognitive capacity may play in different domains of L2 development during intensive language exposure abroad.

To conclude, our key findings that (a) gains in grammatical and lexical sensitivity were experienced after a short-term intensive exposure abroad and (b) these gains were by and large not related to cognitive capacity make important contributions to the study of the context and length of intensive exposure abroad (short-term programs) as well as the type of learners (advanced proficiency). We have taken steps to improve methodology in the field by administering tests *in situ* and ensuring the homogeneity of our sample to the greatest extent possible. Nonetheless, we would like to reiterate the points made by DeKeyser (2014) that

future research should aim to include interactions between variables of interest, such as proficiency level, length of stay, and cognitive capacity, to address which combinations of variables yield the greatest linguistic outcomes during intensive L2 exposure abroad. Such research will serve to refine our understanding of what type of exposure context is most beneficial for certain learners and particular language skills (Collentine & Freed, 2004).

Limitations

Appropriate operationalization of a control group is an important consideration in all behavioral scientific research. Particularly with SA research, it is impossible to randomly assign participants to groups and, as a result, there is a high probability that students who choose to study abroad differ nontrivially from those who do not. SA students are self-selected, and therefore may be more motivated, more independent, have increased language aptitude or cognitive capacity, or otherwise differ in nonnegligible ways from AH comparison groups (DeKeyser, 2014). Especially in the case of summer programs, AH classes tend to be remedial whereas participation in SA programs, at least for our sample, is highly competitive. In addition to these inherent learner-internal differences between SA and AH students, there will often times also be external differences in quality and quantity of L2 input in SA compared to AH programs. For example, professors and assistants involved in AH programs are liable to accommodate nonnative speakers more than those in a SA site, SA programs offer more opportunities for interaction with nonprogram-related speakers, and the cultural and historic fieldwork possible in a SA program cannot be entirely replicated in an AH program. Of course, domestic immersion programs can also yield significant linguistic gains, even outperforming SA programs (e.g., Freed et al., 2004). However, a simple comparison of intensive SA and formal classroom AH programs is, in most cases, an unbalanced comparison and was not appropriate for the aims of our study. Our research goal was to examine the extent of L2 development during 5 weeks of intensive exposure abroad in advanced proficiency L2 learners rather than to compare SA to other exposure contexts. The results of this research lead us to conclude that participants experienced significant gains in L2 grammar and vocabulary during this intensive language exposure abroad, though we cannot make claims as to how those gains

would compare to those of AH learners or learners in domestic immersion programs.

An intensive language program abroad is a naturally complex exposure context. In our study, the program was composed of L2 content classes, fieldwork activities, language conversation partners, and a language pledge. All of these factors contributed to the learners' L2 exposure. As a result, we cannot isolate which subcomponents may have been most responsible for the linguistic gains seen in our participants, and instead consider the program for what it was designed as, an integrated whole that led to significant development in L2 grammar and vocabulary, with almost no observed relationship to cognitive capacity. Future research could disentangle these subcomponents, for example by comparing programs with and without a language pledge. However, such research would require a very large sample size and programs that differ in only one variable, both of which are difficult to reliably obtain in SA research.

Our results are also specific to the nature of the intensive program abroad: advanced L2 learners during a short-term summer program. As noted earlier, a high level of participation in short-term programs warrants research in this area and therefore this may be viewed as more of a strength than a limitation. Additionally, the lack of SLA research on advanced learners (Ortega & Byrnes, 2008) highlights the important contribution this study makes to the field of SLA as well as the subfield of SA research. Finally, the strict entrance requirements may have limited variability amongst students, thus yielding the medium to strong effects in L2 development but potentially leveling out variation in cognitive capacity and reducing the possibility for a significant correlation to emerge with L2 gains. This would most likely be the case with PWM, as the relatively small standard deviations in Table 6 show. However, L1 PWM was the measure of cognitive capacity that did return a significant correlation. In addition, despite greater variation in WM scores, WM failed to produce significant correlations with linguistic gains. It may be the case that having strict entrance requirements is also not truly a limitation, as those strict requirements ensured that wide ranges in proficiency level were not an intervening variable in our study, as they may have been in previous studies using proficiency self-ratings (e.g., Tokowicz et al., 2004) or average number of previous semesters of Spanish instruction (e.g., Sunderman & Kroll, 2009). Furthermore, the findings from our study are ecologically valid considering that a sizable

number of colleges and universities are becoming ever more academically selective.

CONCLUSION

This study demonstrated significant morpho-syntactic and lexical development during a short-term intensive L2 exposure abroad, thereby contributing considerably to the limited research on shorter study abroad programs (Cubillos et al., 2008; Llanes, 2011; Llanes & Muñoz, 2009) and on the effects of SA for advanced proficiency learners (DeKeyser, 2007, 2010, 2014). In addition to those external variables, we also considered the potential relationship between variation in cognitive capacity and L2 gains. We avoided methodological problems by testing participants *in situ*, thereby providing a more accurate perspective of gains and their relationship to cognitive capacity during study abroad. Finally, the finding that individual differences in WM and PWM capacities were not related to L2 development in our participants is informative for language practitioners, program directors, and future study abroad students. The current study has demonstrated that intensive exposure abroad, even for only 5 weeks, is useful for making significant progress in the L2 at an advanced proficiency level. In fact, intensive L2 exposure abroad seems to be a favorable context for linguistic gains at this level, since it appears not to be constrained by differences in cognitive capacity. Rather, the combined effects of the richness of the context derived from extensive input and frequency of interaction led to gains in target language grammar and vocabulary that in large part were not constrained by variation in learners' cognitive resources.

ACKNOWLEDGMENTS

Initial reports of the data were presented at the 2012 American Association for Applied Linguistics in Boston, MA, and the 2012 Georgetown University Roundtable on Languages and Linguistics, Washington, DC. The authors would like to thank the administrators in Georgetown University's Office of International Programs for their support and the students who volunteered many hours of their time in Barcelona to provide us with data, as well as the research assistants who assisted with aspects of data coding. The study is part of The BarSA Project, developed by Cristina Sanz with funding from *Language Learning* and Georgetown University's International Initiatives. We would also like to thank Janire Zalbidea for her insightful

comments on an earlier version of this paper. Any remaining errors are exclusively the responsibility of the authors.

NOTES

¹ We follow VanPatten's (2002) definition of intake: "the linguistic data actually processed from the input and held in working memory for further processing" (p. 757).

² Every Tuesday morning, all day every Thursday, and one weekend during the program students and faculty conducted fieldwork as a group. Fieldwork consisted of visits to sites in Barcelona (Tuesdays) and the broader Catalonia region (Thursdays) that were organized following chronological history, from Roman times to the 1975 democratic transition. Coursework was also organized chronologically, so that there was a direct connection between course content and fieldwork activities. Students listened to the visiting faculty (mostly Ph.D. students from the host institution) that led the visits; students asked questions, took notes, took pictures, and collected brochures or any other free literature for use in the short papers required as part of their coursework.

³ Both the GJT and LDT used in this study were piloted on five native speakers of Spanish prior to their use in the BarSa project.

⁴ The frequency calculations are based on the total noun count in the language; this was done because some of the words can be either nouns or verbs (e.g., hug, 'abrazo') so care was taken to accurately characterize the status of the word within its targeted grammatical class and not to confound these word frequencies with occurrences of a target word as a verb. We are grateful to Dr. Michael Ferreira for his help in tracking down these noun counts and also to Dr. Mark Davies for providing this information from his corpora for a previous project.

⁵ Note that the three measures of aspects of working memory did not correlate with each other, all $ps > .05$, and that this lack of inter-correlations among any of the three measures permitted us to analyze them separately against the difference scores, as opposed to creating a composite z-score of memory capacity from performance on the three tasks.

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